JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 100, NO. B8, PAGES 14,999–15,007, AUGUST 10, 1995

Paleomagnetism of external southern and central Dinarides and northern Albanides: Implications for the Cenozoic activity of the Scutari-Pec transverse zone

Catherine Kissel and Fabio Speranza

Centre des Faibles Radioactivités, CNRS/CEA, Gif-sur-Yvette, France

Vlado Milicevic

Geomagnetic Institute, Grocka, Serbia

Abstract. Paleomagnetic investigations have been conducted on 43 Cretaceous to Miocene sites (500 oriented cores) from the southern and central external Dinarides and from the northern external Albanides. The 24 reliable sites (out of 33) from the Dinarides do not document any significant rotation with respect to Africa since at least early Eocene. This result contrasts with the very coherent post-Eocene 45° clockwise rotation previously observed in the external Albano-Hellenides from Peloponesus to the Tirana basin. These new results suggest that the Scutari-Pec line, located between the two orogens and considered as a Mesozoic transform fault on the basis of geological observations, is the place where the torsion between the Dinarides and the Albanides recently occurred. On a local scale, four sites (out of 10) from two ridges located in northern Albania, south of the Scutari-Pec line but with a Dinaric trend, show that these two structures have undergone a ~30° post-Eocene clockwise rotation. This intermediate value of rotation between the north and the south suggests that the Scutari-Pec line is, in fact, a rather wide transversal zone in which the two ridges are located. It acted as a decoupling zone between the Dinarides and the Albano-Hellenides during the Cenozoic and probably until the Pleistocene time. The strong seismic activity recorded in this region and the anomalously high positive heat flow (documented by other authors) are also evidence for the recent activity of the Scutari-Pec transverse zone.

Introduction

It is now well-established that the geometry of the Mediterranean Alpine belt arises from the Upper Cretaceous-Cenozoic collision between the African and European plates [*Tapponnier*, 1977; *Dercourt et al.*, 1986]. Kinematic reconstructions show that the collision is presently taking place at an average rate of roughly 1 cm/yr [*Argus et al.*, 1989]. Yet in many domains, deformation rates greater by almost an order of magnitude have been documented [*Le Pichon et al.*, 1993]. These high rates result from changing regional trends of the chains, formation of arcs, and recent back arc oceanic basins, which regionally modify the overall pattern of the collision. In this framework, paleomagnetic studies which allow the description of rotation about vertical axis have proven crucial to unravel the complex tectonic pattern and to allow palinspastic reconstruction.

In the central Mediterranean, the Dinarides and Albanides belts are separated by the Scutari-Pec major tectonic discontinuity. South of this boundary, extensive paleomagnetic studies [*Laj et al.*, 1982; *Horner and Freeman*, 1983; *Kissel et al.*, 1985] have revealed consistent clockwise rotation from Peloponnesus to northwestern Greece of about 25° since the lower Pliocene and some 40°-45° since the Eocene. More recently, these results have been extended to Albania, both in the internal [*Mauritsch et al.*, 1991] and external zones [*Speranza et al.*, 1992, 1995]. North of this region, Istria has undergone a poorly defined 30° post-Cretaceous counterclockwise rotation with respect to Africa [*Marton and Veljovic*, 1983], while the Kvarner islands and surroundings yielded Cretaceous paleomagnetic directions consistent with the expected African ones [*Marton et al.*, 1990].

The ages of the formations from which these results were obtained along the northern Dinaric coast on one side and in Albania and northern Greece on the other side are not the same. The differences in the rotational patterns may therefore arise from rotations taking place at different time in the two regions. However, it seems reasonnable to suppose that the northern and the southern regions which have undergone two different rotational behavior are separated by a major decoupling zone. This hypothesis is certainly sustained by the significant change of the trend of the orogens, from N120° in the Dinarides to N150° in the Albano-Hellenides.

Geological Setting and Sampling

The different tectonic units which constitute the Dinarides were thrusted onto the Adriatic foreland during Upper Cretaceous and Lower Cenozoic times [*Aubouin et al.*, 1970;

Copyright 1995 by the American Geophysical Union.

Paper number 95JB01243
0148-0227/95/95JB-01243$05.00

Aubouin, 1973]. The most external units are from east to west: the Karst, the Budva, and the Dalmate Zones (Figure 1).

The Karst Zone (Albanian Alps in Albania) is characterized by a thick Mesozoic calcareous sequence, covered by thin deposits of upper Cretaceous-middle Eocene flysch. The flysch sequence predates the main tectonic phase, which has affected the Karst during middle Eocene. During this major event, the rigid calcareous shelf of the Karst Zone split into several tectonic slices which were thrusted southeastward onto the more external zones. After complete emersion during the Eocene, several subsident basins developed in the Karst unit and were filled by Oligocene to Pleistocene lacustrine sediments.

The Budva Zone (Krasta-Cukali in Albania) crops out only in the southern Dinarides. It disappears about 20 km south of Dubrovnik where the Karst directly overthrusts the Dalmate zone (Figure 1). The sedimentary sequence of the Budva Zone remained entirely pelagic from Trias to upper Cretaceous. It became detritic during the uppermost Cretaceous (Maastrichtian) up to the Eocene when the flysch started to fill the preexisting basins.

The Dalmate Zone (Kruja Zone in Albania) crops out in a narrow band all along the Adriatic coast and in all the eastern Adriatic islands (Figure 1). The Mesozoic sequence is entirely made of shelf limestones, very similar to the ones that outcrop in the Karst. For this reason the boundary between the Karst Zone and the Dalmate Zone is not clear in central and northern Dinarides. The calcareous sedimentation extends in the Dalmate Zone up to middle Eocene and is followed by middle-upper Eocene flysch. At the Eocene-Oligocene boundary the deformation reached the Dalmate Zone, which was divided in several tectonic units overthrusted onto the Adriatic foreland. The age of the compressive phase affecting the Kruja Zone (and also the southernmost part of the Dalmate Zone south of Bar [*Radoicic et al.* ,1989]) is significantly different from the age of the deformation reported in the rest of the Dalmate Zone from Kotor in the south to the north. In Albania and in the southernmost Dinarides, the calcareous sedimentation extended during all the Eocene period and ended with the Oligocene-Aquitanian flysch. This sequence was deformed during lower Miocene (presumably during upper Aquitanian).

The Dinarides are separated from the Albanides by a major tectonic feature: the Scutari-Pec line, considered for a long time as an ancient Mesozoic transform fault affecting the Adriatic passive margin (Figure 1). The disappearing of the Karst Zone in the south and the ~100 km SW displacement of the ophiolitic nappe in Albania indeed suggest that the Scutari-Pec zone was initially a paleogeographic boundary and is evidence for old activity [*Aubouin and Dercourt,* 1975]. On the other hand, the most external zones of the orogen (Budva (Krasta Cukali) Zone and Dalmate (Kruja) Zone) are not shifted by this tectonic feature, although their trend changes of about 30° from the Dinarides to the Albano-Hellenides.

Figure 1 shows schematically the geographical position of the sampling areas distributed mainly in the most external zones of the central southern Dinarides and of the northern Albanides. Out of the 43 sampled sites, 24 correspond to fine-grained blue-grey marls of the flysch sections, and 19 correspond to marine and continental limestones. The flysch sections are mainly located along the Adriatic coast in the middle to late Eocene sections of the central Dalmate zone between Split and Makarska (19 sites) (Figure 2a). The Split area is characterized by a progressive change in the strike of regional scale folds. The eastern limb of the deflection (along the coast) trends N120°, parallel to the main structural features of the Dinaric belt. In the central part of the Split deflection

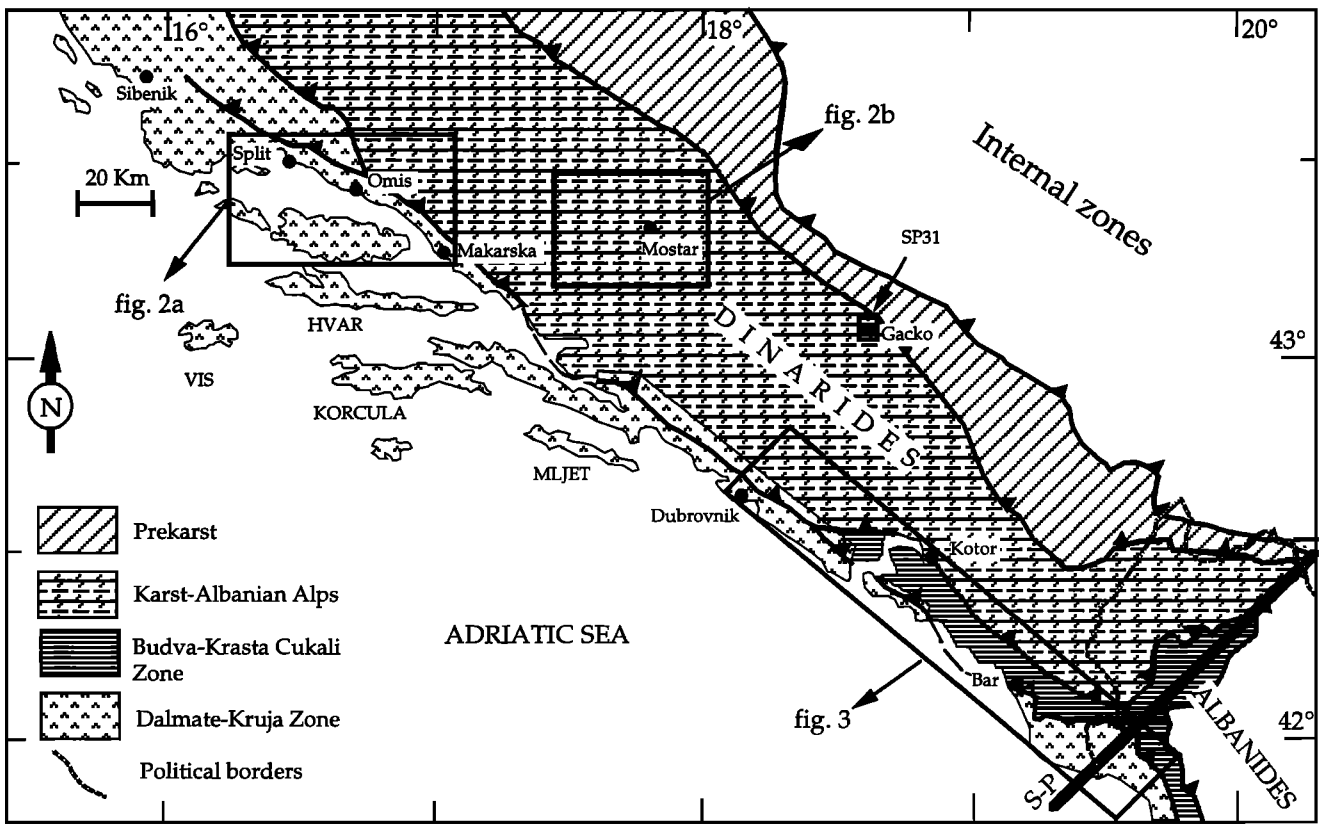


Figure 1. Structural map of Dinarides and northern Albanides, showing the location of the studied areas. S-P is Scutari-Pec transverse zone.

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

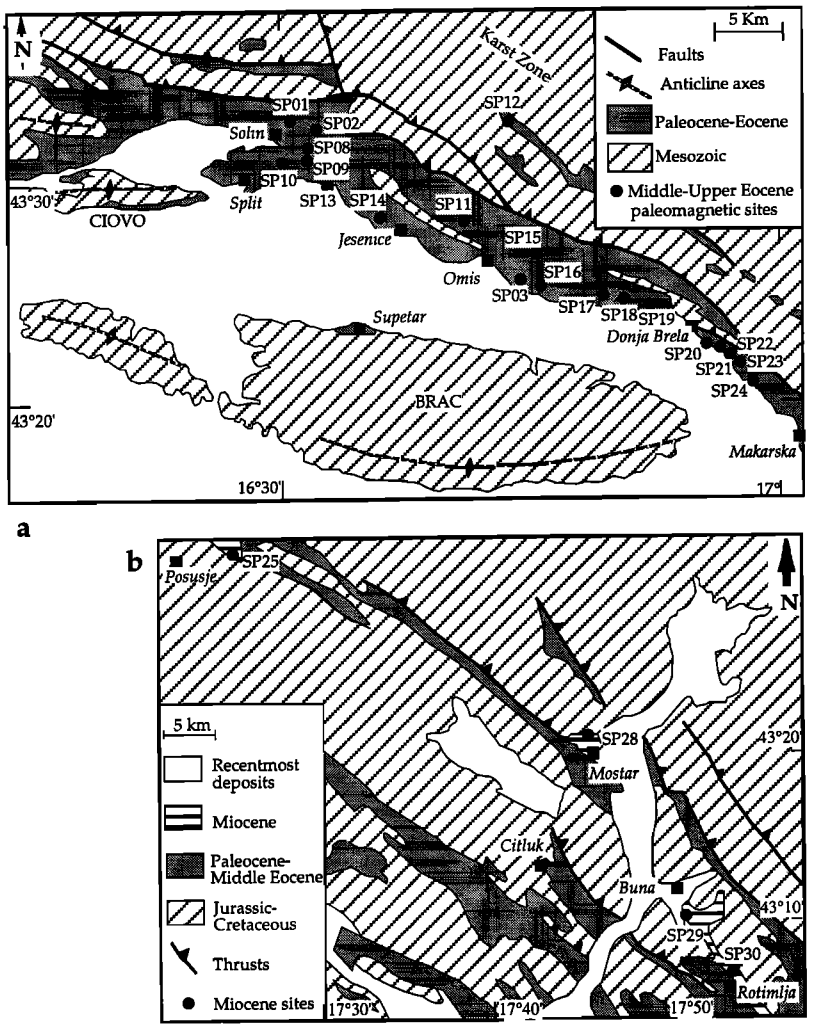


Figure 2. Geological scheme of the (a) Split area and (b) Mostar area and location of paleomagnetic sampling sites.

the trend of the structures is E-W as it can be seen in the islands of Korcula, Brac, Vis, and Hvar (Figure 1). The western part of the deflection is obscured by the Adriatic Sea, but it recovers a clear N120° trend at the latitude of Sibenik. Our sampling sites are distributed along the eastern limb of the deflection within folds and minor thrust sheets, the orientation of which varies from a regional N120° trend (for the majority of them) to a local E-W trend. Site SP12, which is geographically close to the others, belongs to the Karst Zone (Figure 2a).

Four Miocene calcareous sites were sampled in continental calcarenites from the Karst Zone, near Mostar (Figure 2b). These limestones form small patches which unconformably cover older marine series and seal middle Eocene thrust faults. In the same region we also sampled one site (SP31) in the upper Miocene-Pliocene lacustrine clays of the Gacko basin, about 60 km SE of Mostar (Figure 1). This small basin, discordant on Mesozoic limestones, contains a presently exploited coal lens.

Farther south, the flysch sections of the Sutorina valley monocline, which extends from Kotor to Dubrovnik, were also sampled (sites SP35, SP40, and SP41 in Figure 3). In this monocline, which follows the regional Dinaric direction, five sites (SP32, SP33, SP34, SP36, and SP37) were sampled in lagunal and marine limestones which conformably underlie the flysch. The ages of these sediments range from early to late Eocene [*Cadet,* 1978].

The last 10 calcareous sites were sampled in Eocene (six sites) and Cretaceous (four sites) shelf limestones from the Kruja (Dalmate) Zone, in northern Albanides (Figure 3). These limestones form two calcareous ridges (Rencit and Kakarriqit ridges) exhibiting a "Dinaric" trend (N120°), south of the western extremity of the Scutari-Pec tectonic line as it has been described in the litterature.

All the samples were drilled with an electric portable drill supplied with a power generator and were oriented with standard paleomagnetic techniques using both a magnetic and a sun compass. Eight to 17 cores were drilled at each site (11-12 as an average), and each core yielded two to five specimens for directional and mineralogical paleomagnetic analyses.

Results of the Laboratory Experiments

Rock Magnetic Characteristics

All the measurements were made in a shielded room with a 2G three-axis cryogenic magnetometer. The natural remanent magnetization (NRM) of all the specimens from 14 sites (six marly, eight calcareous) was found too weak (less than 10^{-5} A/m at room temperature) to be measured with the necessary

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

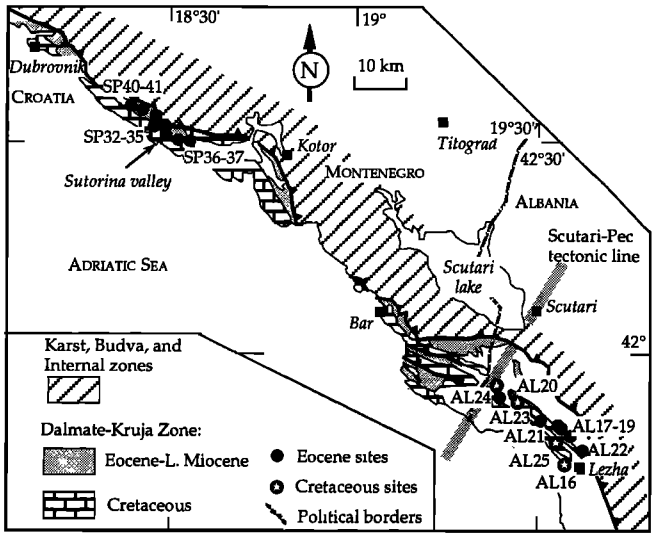


Figure 3. Geological scheme of the southern Dalmate zone-northern Albanides margin and location of paleomagnetic sampling sites.

accuracy or proved totally unstable upon thermal or alternating field demagnetization. After checking that the low NRM did not result from compensation of magnetizations of opposite sign, these 14 sites were not investigated further. The average NRM of the other sites is about $5x10^{-4}$ A/m for the marly samples and $1x10^{-4}$ A/m for the limestones.

Stepwise acquisition of the isothermal remanent magnetization (IRM) was conducted on at least one sample per site. The marly samples show an almost complete saturation between 0.2 and 0.3 T and a remanent coercivity Hcr between 40 and 60 mT (Figure 4a). Therefore, in these samples, no high-coercivity minerals contribute to the magnetization. The unblocking temperature spectra of the IRM were analyzed using the method suggested by *Lowrie* [1990] and clearly show that the soft fraction is completely removed between 500° and 600°C (Figure 4b). These experiments indicate that in all the marly samples, the remanent magnetization is carried by pure magnetite or by moderate Ti content magnetites.

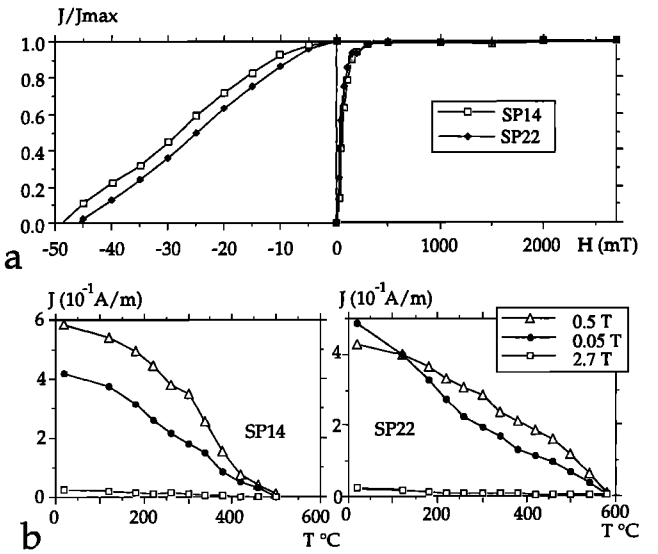


Figure 4. Typical magnetic properties of marly samples. (a) Stepwise acquisition of the isothermal remanent magnetization (IRM) and determination of the remanent coercivity (Hcr). (b) Thermal demagnetisation of a three-component IRM produced by magnetizing the samples in 2.7, 0.5, and 0.05 T successively along the three different sample axes. All these analyses show that magnetite is the main magnetite carrier in the marly formations.

Concerning the limestones, for six out of the 11 sites investigated, a complete saturation is acquired before 0.3 T (Figure 5a). For the other five sites (the four sites from northern Albanides and site SP30 from central Dinarides), the remanent magnetization at 0.3 T represents 70-90% of the maximum magnetization at 2.7 T, although saturation has not been reached by this field. The low-coercivity fraction being predominant, the Hcr always ranges between 40 and 60 mT, as for the marly samples (Figure 5a). In all the samples the soft fraction is entirely demagnetized between 450° and 600°C (Figure 5b). This implies that magnetite (sometimes with some titanium content) is the only magnetic carrier in six calcareous sites and the main carrier in the other five sites. In these five sites the high-coercivity component is completely removed only between 650° and 700°C and is thus carried by hematite (Figure 5b).

Finally, the anisotropy of the low-field magnetic susceptibility has been systematically investigated using a Kappabridge KLY-2. Except for three sites which clearly exhibit a prolate fabric of tectonic origin, we always observe an oblate susceptibility ellipsoid (2.6% flattening) with a minimum axis perpendicular to the bedding indicating the sedimentary origin of the fabric. A magnetic lineation is also well marked in these sites by a clustering of the maximum susceptibility axes (1.6% lineation). According to previous results obtained from the same kind of sediments in neighboring regions [*Kissel et al.*, 1985], this lineations indicate that the magnetic fabric can be partly of tectonic origin, in agreement with the prolate fabric observed in the other sites.

Paleomagnetic Directions

One specimen per site has been stepwise demagnetised using 10-15 steps from room temperature up to the limit of reproducible results (450°-600°C). In the great majority of the marly samples, apart from a viscous component which is removed at 200°C, the NRM decreases continuously and the

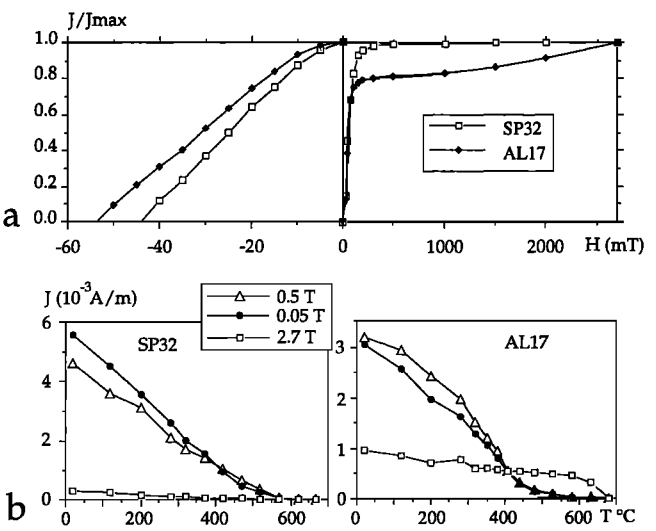


Figure 5. Typical magnetic properties of calcareous samples (same experiments as in Figure 4). In six of the sites (example of site SP32), magnetite is the only magnetic carrier, while in the five other sites (example of site AL17), hematite is also present.

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

direction of magnetization remains stable (Figure 6). The low-field magnetic susceptibility measured at each step generally increases by a factor of 2 to 6 between 400° and 500°C, indicating that mineralogical changes occurred. These changes certainly correspond to the acquisition of a very weak magnetization only visible in the less magnetized samples which show an erratic behavior of the directions of magnetization after 400°C. In the calcareous samples, no major change in the low-field magnetic susceptibility was observed during heating, and thus no mineralogical changes were detected. In the Eocene limestones from northern Albanides, NRM intensities are very weak (1 to 3 10^{-5} A/m), so that the direction of the stable component is less precisely defined (sample AL22 10A in Figure 6b). In about one third of the calcareous and marly samples, two components of magnetization still coexist after removal of the viscous component and in this case, we have calculated the best great circle fitting the NRM directions (Figure 6c).

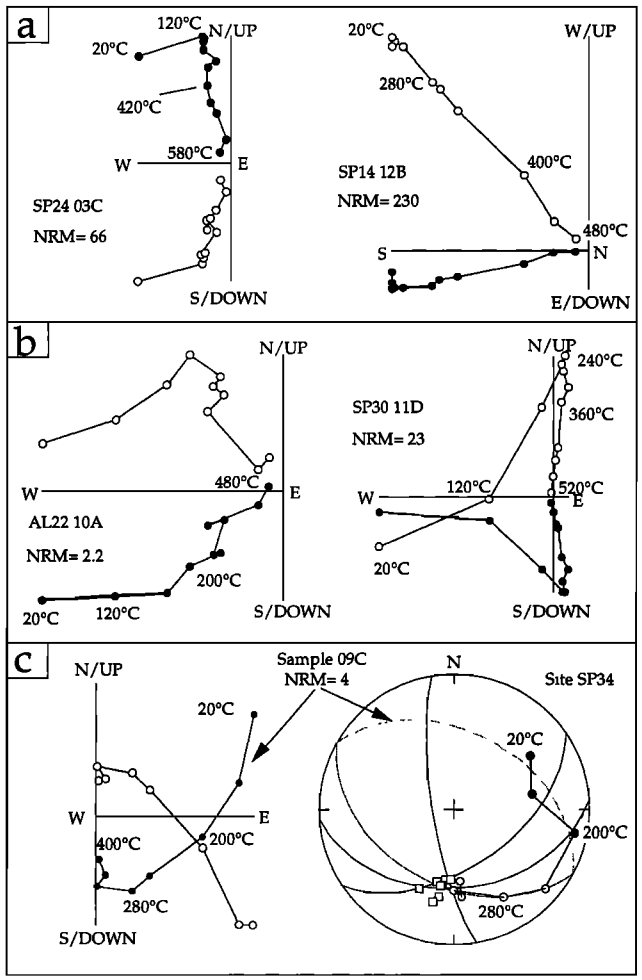


Figure 6. Typical thermal demagnetization diagrams after bedding correction of (a) the marly and (b) the calcareous samples. Solid and open circles correspond to the projections onto the horizontal and vertical planes respectively. (c) Example of demagnetization diagram of a sample in which two components of magnetization are simultaneously demagnetized. The stereographic projection of the great circle corresponding to this sample is also shown together with the stable directions (squares) and the other great circles obtained from the same site. The intensity of the natural remanent magnetization (NRM) is reported in 10^{-5}A/m.

Site-mean directions were calculated using *Fisher*'s [1953] statistics when only stable directions were obtained or combining stable directions and great circles using *McFadden and McElhinny*'s [1988] method. The site-mean directions from the Dinarides, based on 5 to 13 directions, are precisely defined with α_{95} ranging between 2.5 and 10 (only the α_{95} obtained for site SP09 is greater) (Table 1a). The slightly higher values of α_{95} in the four reliable sites from northern Albanides can be explained by the very weak magnetization of the samples which did not allow a very precise determination of the direction of the characteristic component.

The reliable results are reported in Tables 1a and 1b in four groups depending on the location and the age of the sites: middle-upper Eocene in the Split-Makarska area (central Dinarides), Eocene in the Sutorina valley (southern Dinarides), Eocene in the northern Albanides, and Mio-Pliocene in the Karst Zone (central Dinarides). After bedding correction, normal and reverse directions from Eocene sites are perfectly antipodal and confirm the efficiency of the demagnetization process (Figure 7). The Miocene sites from the Karst zone show only reverse directions except site SP28 which exhibits a mean direction close to the present-day field direction before bedding correction (D=351°, I=62°). This direction is strongly deviated from the other Miocene directions after bedding correction, and we think that this site has been recently remagnetized, although mineralogical analyses have demonstrated that magnetite is the only magnetic carrier. A similar situation has been recently described in Eocene porous limestones from southern Italy, with a lithology very similar to the one of site SP28 [*Speranza and Kissel,* 1993]. We did not consider this site in the calculation of the regional mean direction.

The statistical parameters associated to the mean regional directions and reported for each group of sites in Table 1b, clearly show a significant decrease (1% significance level) of the scatter of the data upon tectonic correction (Table 2). The example of the Eocene formations in Split-Makarska area is given in Figures 8a and 8b, while the numerical values are reported in Table 2. These positive tilt tests demonstrate the primary character of the magnetization as the tectonic deformations occurred just after the deposition of the studied sediments.

In the Split-Makarska region, although the fold test is positive, we observe that after tectonic correction, the mean declinations are slightly more scattered than the inclinations (Figure 8b). This may result from slight local changes in the orientation of the fold axes due to the Split deflection. In order to estimate the effect of this structural scatter on the paleomagnetic directions, we corrected the declinations for deviation of the fold axis orientation by aligning the fold axes with the regional trend (N120°). The strike of the structures, especially for sites showing an horizontal bedding plane, was evaluated using the magnetic lineation which has developed perpendicular to the shortening direction. The changes in the structural directions are not large enough to allow a meaningfull numerical test of the oroclinal bending [*Schwartz and Van der Voo,* 1983]. However, although the correction for the structural axes only slightly changes the mean declination value, it improves the grouping of the data with a K parameter increasing from 49 after a simple tilt correction to 65 after this additional structural correction (Figure 8c and Table1b). This result supports the interpretation that the paleomagnetic directions predate the formation of the local deflection.

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1a. Site-mean Paleomagnetic Directions From Dinarides and Albanides

Site	$(n_1,n_2)/n_3$	D_b,deg	I_b,deg	D_a,deg	I_a,deg	K	α_{95},deg	D_{ac},deg
			Central Dinarides : Split-Makarska Area (Middle-Upper Eocene)					
SP02	(8,0)/12	14	64	4	49	57	7.4	4
SP03	(11,0)/14	359	67	15	39	36	7.7	18
SP08	(8,0)/10	353	36	339	48	37	9.3	346
SP09	(7,0)/8	325	61	334	35	21	13.5	5
SP12	(11,0)/11	346	43	349	35	107	4.4	4
SP13	(9,3)/12	324	59	353	50	70	5.3	355
SP14	(12,0)/12	195	-39	169	-51	267	2.7	176
SP16	(8,2)/11	282	56	332	49	131	4.3	355
SP17	(6,1)/10	349	-29	341	49	92	6.4	358
SP19	(7,0)/16	347	-28	345	42	50	8.6	21
SP20	(1,6)/9	258	66	358	60	142	5.8	7
SP21	(3,2)/11	316	57	3	36	66	10.0	14
SP22	(5,6)/11	292	70	355	47	45	7.1	355
SP24	(9,1)/10	317	61	5	41	91	5.1	5
			Southern Dinarides: Sutorina Valley (Eocene)					
SP32	(4,7)/12	167	-71	180	-44	355	2.5	
SP33	(1,5)/12	148	-68	186	-42	98	8.0	
SP34	(8,4)/12	173	-66	191	-41	140	3.7	
SP35	(0,12)/14	291	59	8	40	87	4.7	
SP40	(3,5)/10	356	65	8	38	63	7.5	
SP41	(1,11)/12	71	-64	196	-42	49	6.7	
			Northern Albanides (Eocene)					
AL17	(2,6)/11	342	76	25	36	38	9.9	
AL19	(6,2)/10	356	64	30	39	13	16.0	
AL21	(1,7)/10	220	19	219	-45	25	12.0	
AL22	(7,2)/12	218	8	209	-63	29	9.2	
			Central Dinarides: Karst Zone (Mio-Pliocene)					
SP25	(4,2)/10	194	-47	187	-46	90	7.4	
SP28 *	(13,0)/17	351	62	318	59	116	3.9	
SP29	(10,0)/11	40	-78	196	-56	341	2.6	
SP30	(7,2)/11	194	-62	172	-55	120	4.8	
SP31	(12,0)/13	178	-52	181	-46	193	3.1	

Here $(n_1,n_2)/n_3$ is (number of stable directions, number of great circles)/total number of studied samples in the site. The differences between (n_1,n_2) and n_3 correspond to the number of non reliable samples in each sites (magnetically unstable or deviated from the main group by more than 2σ). Declination and inclination are calculated before (D_b, I_b) and after (D_a, I_a) bedding correction was applied. D_{ac} is the declination obtained after orocline correction (see text). α_{95} and K are the parameters of the Fisher's statistics.

*: Remagnetized site (see text).

Table 1b. Regional Mean Paleomagnetic Directions From Dinarides and Albanides

	n/N	*D*, deg	*I*, deg	*K*	α_{95}, deg
		Central Dinarides : Split-Makarska Area (Middle-Upper Eocene)			
Mean before bedding correction:	14/20	325	71	5	20
Mean after bedding correction:	14/20	352	46	49	5.7
Mean after orocline correction:	14/20	4	45	65	5.0
		Southern Dinarides: Sutorina Valley (Eocene)			
Mean before bedding correction:	6/8	141	-70	24	14
Mean after bedding correction:	6/8	188	-41	334	3.7
		Northern Albanides (Eocene)			
Mean before bedding correction:	4/10	238	56	2.5	74
Mean after bedding correction:	4/10	31	46	41	14
		Central Dinarides: Karst Zone (Mio-Pliocene)			
Mean before bedding correction:	4/5	185	-65	11	29
Mean after bedding correction:	4/5	184	-51	98	9.3

Here *n/N* is number of sites taken into account in the statistical calculations/total number of studied sites. *D* is declination; *I* is inclination; *K* and α_{95} are the parameters of the Fisher's statistics

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

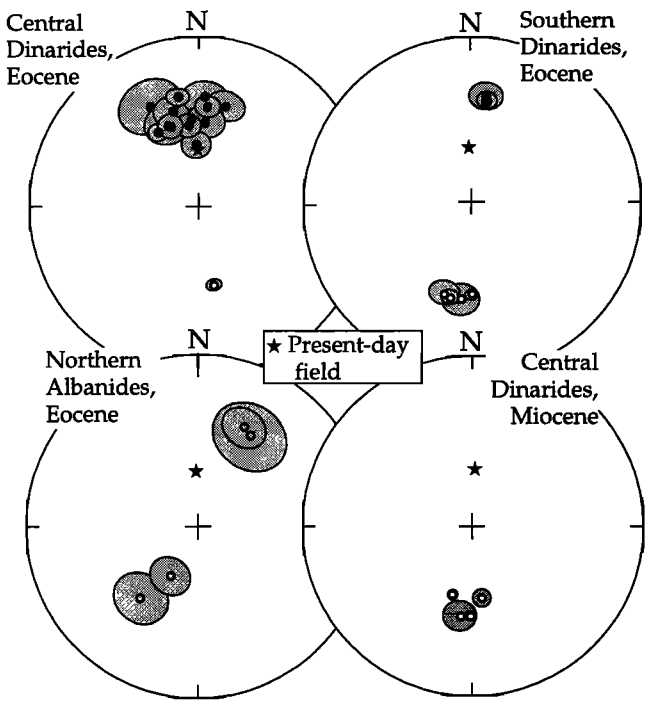


Figure 7. Equal-area projections of the site-mean directions of magnetisation after bedding correction obtained from the Dinarides and the northern Albanides, with associated 95% confidence circles. Solid and open circles correspond to downward and upward directions, respectively.

Discussion

Because no Cenozoic poles are available for the Adriatic region, we have referred our data to the African poles proposed in the most recent compilation [*Besse and Courtillot,* 1991]. Rotation (R) and flattening (F) parameters were calculated with respect to Africa using the equations given by *Beck* [1980] and modified by *Demarest* [1983].

Flattening is not significant, except for the upper Eocene flysch units from the Sutorina valley (12°± 6°). These units are very rich in clay minerals, and the observed flattening is certainly due to a compaction effect as it has been already proposed in similar lithologies [*Laj et al.,* 1982]. Except for these three sites, both Eocene and Mio-Pliocene inclinations obtained from the studied area are coherent with the African expected ones.

The mean regional declinations obtained from the Split-Makarska area (central Dinarides), from the Sutorina valley (southern Dinarides), and from the Neogene basins in the Karst Zone (central Dinarides) are very similar to the African

Table 2. Positive Tilt Tests Obtained From the Cenozoic Formations in the External Dinarides and the External Albanides

Geographic Position	Age	N	K_2/K_1	$F_{1\%}$
Split-Makarska area	Eocene	14	9.6	2.5
Sutorina valley	Eocene	6	14	4.8
Northern Albanides	Eocene	4	16	8.5
Karst Zone	Mio-Pliocene	4	8.9	8.5

N is number of sites. The tests are positive at 1% confidence level when $K_2/K_1 > F_{1\%}$ (K_2 and K_1 are the dispersion parameter of the Fisher's statistic before and after bedding correction, respectively).

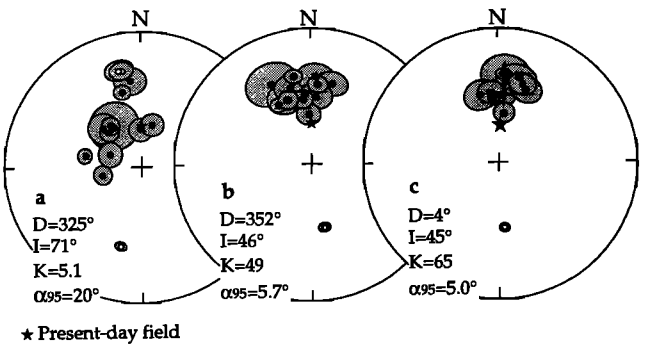


Figure 8. Equal-area projections of the site-mean directions (with 95% confidence circles) obtained from the Split-Makarska area: (a) before tilt correction; (b) after tilt correction; and (c) after tilt and orocline correction. The scatter of the data decreases significantly upon tilt and orocline corrections (see text). Symbols are the same as in Figure 7.

expected ones, as it is shown by the negligible values of the R parameters (Table 3). This null rotation with respect to Africa is consistent with the results obtained from the Cretaceous formations in the external northern Dinarides [*Marton et al.,* 1990), from the Permo-Triassic to Cretaceous formations in the southern Alps [*Channell et al.,* 1992; *Channell and Doglioni,* 1994; *Channell,* 1994], and from upper Cretaceous and Eocene limestones in the Gargano peninsula [*VandenBerg* 1983 revised by *Van der Voo,* 1993; *Speranza and Kissel,* 1993]. It suggests that these peri-Adriatic regions did not undergone any significant rotation with respect to Africa since at least lower Eocene time.

The null rotation observed in the Dalmate zone is clearly different from the post-Eocene 45° clockwise rotation observed very consistently in Albania [*Speranza et al.,* 1995] and in northwestern Greece [*Horner and Freeman,* 1983; *Kissel et al.,* 1985]. This implies the existence of a huge decoupling zone between these two studied areas which are very close to each other (about 100 km). The only obvious tectonic discontinuity which can play this role between the Dinarides in the north and the Albano-Hellenides in the south is the Scutari-Pec transverse line.

Let us consider now the results we obtained from the two ridges in Northern Albania which were supposed to be located south of the Scutari-Pec line although they have a Dinaric trend. The Eocene sites sampled in these two structures yield a 30° clockwise rotation with respect to Africa (Table 3). This post-Eocene rotation is larger than the one observed in coeval formations in the Dinarides and about 15° smaller than in the south from Peloponnesus to the basin of Tirana in Albania. It suggests that the Scutari-Pec line is, in fact, a rather wide transversal zone in which the two ridges are included. The rotation of the two northern Albanian ridges can be interpreted either in terms of intermediate rotation in an oroclinal bending or in terms of block rotation that one could expect in such a hinge region.

The Scutari-Pec transversal zone is clearly the northern limit of the region involved in the clockwise rotation and thus marks the northwestern limit of the Aegean system. As a first approximation, we can consider that the rotation observed in the Albano-Hellenides is a semirigid one, as suggested by the remarkable consistency of the paleomagnetic results over the belt and the absence of large-scale strike-slip faults. In this case, the pivot point of the clockwise rotation is certainly

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Rotation and Flattening Parameters of the Southern External Dinaric Belt With Respect to Africa

Region	Formation	Age	*N*	*F*, deg	*R*, deg
Split-Makarska	flysch	Eocene	14	6±5	2±7
Sutorina Valley	limestones	Early Eocene	3	5±6	5±5
	flysch	Late Eocene	3	12±6	
North Albanides	limestones	Eocene	4	4±12	28±17
Karst Zone	calcarenites	Mio-Pliocene	4	6±7	-1±12

N: is number of reliable sites; *R, F* are rotation and flattening parameters calculated for each period using the equations given by *Beck* [1980] and *Demarest* [1983]. The African directions were calculated using the African apparent polar wander path proposed by *Besse and Courtillot* [1991].

located at the Scutari-Pec transverse zone, immediatly north of the rotating region. This proximal location is sustained by the absence of horizontal displacement of the external Albanides with respect to the external Dinarides. Moreover, it has been shown that both the basement and the nappes are involved in the Albano-Hellenic rotation [*Speranza et al.*, 1992, 1995]. It implies that the tectonic deformation within the Scutari-Pec transversal zone certainly affects the deep crust.

These new Dinaric results together with data obtained previously from the Albano-Hellenides allow us to reconstruct the Cenozoic evolution of the orientation of the structures in the two belts and the role of the Scutari-Pec transversal zone (Figure 9). The present-day change in the trend of the orogens on both sides of the decoupling zone corresponds to a 25° clockwise rotation of the Albano-Hellenic structures with respect to the Dinaric ones. This apparent torsion of the structures results certainly from the youngest phase of rotation which affected the Albano-Hellenic belt during Plio-Pleistocene [*Laj et al.*, 1982; *Speranza et al.*, 1995]. This suggests that during upper Miocene period (about 12 to 5 Ma), the two belts were almost perfectly aligned along a N120°-125° direction. Before this period, they were diverging by an angle of about 25° in the opposite sense (Figure 9). The Plio-Pleistocene age of the recent rotation of the Albano-Hellenic margin indicates that the decoupling zone has been very recently active. No argument is so far available to give a precise age to the post-Eocene rotation of the two North-Albanian ridges located within the Scutari-Pec transverse zone (it may have occurred during the lower to middle Miocene phase or during the Plio-Pleistocene one or during both phases).

These new data thus bring the first experimental proof that the Scutari-Pec line, considered as a Mesozoic tectonic boundary on the basis of geological observations, is, in fact, a very important tectonic feature during the Cenozoic and probably until the late Pleistocene time acted as a wide and deep decoupling zone between two major orogens of the central Mediterranean.

No obvious tectonic feature is known in the Adriatic offshore facing the Scutari-Pec zone [*Finetti*, 1982], even if this region seems to correspond to an abrupt change in the deflection of the Adriatic lithosphere beneath the eastern Adriatic chains [*Moretti and Royden* ,1988]. However, this area is characterized by a well-marked heat flow anomaly [*Cermak et al.*, 1994], possibly related to the recent tectonic activity of the Scutari-Pec zone. In the same way, the intense

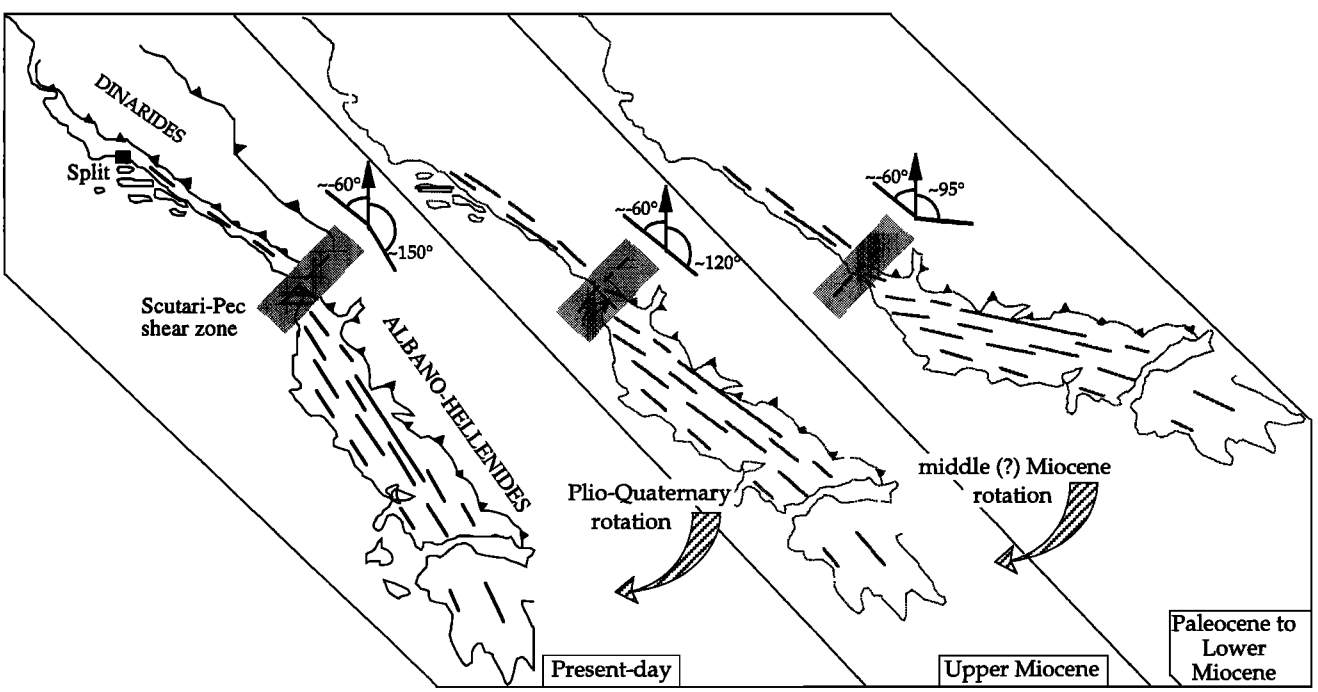


Figure 9. Schematic evolution of the orientation of the external Albano-Hellenic and Dinaric structures during Cenozoic time deduced from the paleomagnetic data. From Paleocene to upper Miocene time, the coastal outlines and the structures are drawn as dashed lines as there are today only to supply a visual support to the rotation.

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

historical seismicity recorded in this area, including the 1979 Montenegro earthquake, is evidence for the most recent activity of the Scutari-Pec transversal zone. Fault plane solution of the earthquakes recorded in the region show mainly thrust mechanisms along faults parallel to the belt without any major strike-slip activity [*Anderson and Jackson,* 1987]. This is in agreement with the fact that the external zones are not shifted by the discontinuity but only distorted by the different rotational patterns.

By carefully distributing the paleomagnetic sites along the eastern Adriatic margin, we could show how the activity of the Scutari-Pec transversal zone has been critical in the evolution of central Mediterranean. This specific zone is still poorly known from a tectonic point of view, and additional geological and geophysical investigations both on-shore and off-shore are now needed.

Acknowledgments. We wish to thank Carlo Laj for having initiated this study and for invaluable discussions. The constructive comments of A. Hirt, J. Channell, and an anonymous referee helped in the improvement of the manuscript. We are grateful to J. Surmont for his valuable assistance with field sampling in Croatia and to I. Islami, S. Pellumbi, and A. Hyseni for their participation in Albania. We also thank I. Gusic and S. Marincic from the Institute of Geology in Zagreb [Croatia] for guiding us in the field in the Split area. The Geomagnetic Institute in Grocka, the Institute of Geology in Zagreb, and the University of Tirana kindly provided working permits. This field work has been supported by the Atomic Energy Commission. F.S. acknowledges a grant from the Atomic Energy Commission. This is a CFR contribution 1705.

References

Anderson, H., and J. Jackson, Active tectonics of the Adriatic region, *Geophys. J. R. Astron. Soc., 91,* 937-983, 1987.

Argus, D. F., R. G. Gordon, C. DeMets, and S. Stein, Closure of the Africa-Eurasia-North America plate motion circuit and tectonics of the Gloria Fault, *J. Geophys. Res., 94,* 5585-5602, 1989.

Aubouin, J., Des tectoniques superposées et de leur signification par rapport aux modèles géophysiques. L'exemple des Dinarides: Paléotectonique, tectonique, tarditectonique, néotectonique, *Bull. Soc. Geol. Fr., 15*(7), 426-460, 1973.

Aubouin, J., and J. Dercourt, Les transversales dinariques, dérivent-elles de paléofailles transformantes?, *C. R. Acad. Sci., D, 281,* 347-350, 1975.

Aubouin, J., R. Blanchet, J. P. Cadet, P. Celet, J. Charvet, J. Chorowicz, M. Cousin, and J. P. Rampnoux, Essai sur la géologie des Dinarides, *Bull. Soc. Geol. Fr., 7*(XII), 1060-1093, 1970.

Beck, M. E., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res., 85,* 7115-7131, 1980.

Besse, J., and V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma, *J. Geophys. Res., 96,* 4029-4050, 1991.

Cadet, J. P., Essai sur l'évolution alpine d'une paléomarge continentale: Les confins de la Bosnie-Herzégovine et du Montenegro (Yougoslavie), *Mem. Soc. Geol. Fr., 133*(LVII), 1-84, 1978.

Cermak, V., M. Kresl, L. Kucerova, A. Frasheri, N. Kapedani, R. Lico, and V. Cano, Heat flow data from Albania, paper presented at the XIX General Assembly of the European Geophysical Society, Grenoble, France, April 25-29, 1994.

Channell, J.E.T, Paleomagnetism, tectonics and paleogeography of Adria, paper presented at a meeting organized by the Geomagnetism and Aeronomy Group of the Joint Association for Geophysics "Paleomagnetism in the Mediterranean Region", London, 1994.

Channell, J. E. T., and C. Doglioni, Early Triassic paleomagnetic data from the Dolomites (Italy), *Tectonics, 13*(1), 157-166, 1994.

Channell, J. E. T., C. Doglioni, and J. S. Stoner, Jurassic and Cretaceous paleomagnetic data from the Southern Alps (Italy), *Tectonics, 11*(4), 811-822, 1992.

Demarest, H. H., Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. Geophys. Res., 88,* 4321-4328, 1983.

Dercourt, J., et al., Geological evolution of the Tethys belt from the Atlantic to the Pamir since the Lias, *Tectonophysics, 123,* 241-315, 1986.

Finetti, I., Structure, stratigraphy and evolution of central Mediterranean, *Boll. Geof. Teor. Appl., 24*(96), 247-312, 1982.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, 217,* 195-305, 1953.

Horner, F., and R. Freeman, Palaeomagnetic evidence from pelagic limestones for clockwise rotation of the Ionian zone, western Greece, *Tectonophysics, 98,* 11-27, 1983.

Kissel, C., C. Laj, and C. Müller, Tertiary geodynamical evolution of northwestern Greece: Paleomagnetic results, *Earth Planet. Sci. Lett., 72,* 190-204, 1985.

Laj, C., M. Jamet, D. Sorel, and J. P. Valente, First paleomagnetic results from Mio-Pliocene series of the Hellenic sedimentary arc, *Tectonophysics, 86,* 45-67, 1982.

Le Pichon, X., N. Chamot-Rooke, P. Huchon, and P. Luxey, Implications des nouvelles mesures de géodésie spatiale en Grèce et en Turquie sur l'extrusion latérale de l'Anatolie et de l'Egée, *C. R. Acad. Sci., Ser. II, 316,* 983-990, 1993.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett., 17*(2), 159-162, 1990.

Marton, E., and D. Veljovic, Paleomagnetism of the Istria Peninsula, Yugoslavia, *Tectonophysics, 91,* 73-87, 1983.

Marton, E., V. Milicevic, and D. Veljovic, Paleomagnetism of the Kvarner islands, Yugoslavia, *Phys. Earth Planet. Inter., 62,* 70-81, 1990.

Mauritsch, H. J., R. Scholger, and P. Alikay, Paleomagnetic results of Albania (U. Cretaceous-Eocene) and their tectonic significance in the frame of the Alpine-Mediterranean tectonic evolution, paper presented at the IAGA meeting for the XX General Assembly of the IUGG, Vienna, Aug. 11-24, 1991.

McFadden, P. L., and M. W. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth Planet. Sci. Lett., 87,* 161-172, 1988.

Moretti, I., and L. Royden, Deflection, gravity anomalies and tectonics of doubly subducted continental lithosphere: Adriatic and Ionian Seas, *Tectonics, 7*(4), 875-893, 1988.

Radoicic, R., P. De Capoa, and B. D'Argenio, Late rather than Early Tertiary deformation of External Dinarides: Stratigraphic evidence from Montenegro, *Rend. Acad. Sci. Fis. Mat. Naples, 56,* 41-49, 1989.

Schwartz, S. Y., and R. Van der Voo, Paleomagnetic evaluation of the orocline hypothesis in the central and southern Appalachians, *Geophys. Res. Lett., 10,* 505-508, 1983.

Speranza, F., and C. Kissel, First paleomagnetism of Eocene rocks from Gargano: Widespread overprint or non rotation?, *Geophys. Res. Lett., 20,* 2627-2630, 1993.

Speranza, F., C. Kissel, I. Islami, A. Hyseni, and C. Laj, First paleomagnetic evidence for rotation of the Ionian zone of Albania, *Geophys. Res. Lett., 19,* 697-700, 1992.

Speranza, F., I. Islami, C. Kissel, and A. Hyseni, Cenozoic geodynamical evolution of the external Albanides: a paleomagnetic approach, *Earth Planet. Sci. Lett., 129,* 121-134, 1995.

Tapponnier, P., Evolution tectonique du système alpin en Méditerranée: poinçonnement et écrasement rigide-plastique, *Bull. Soc. Géol. Fr., 7,* 437-460, 1977.

VandenBerg, J., Reappraisal of paleomagnetic data from Gargano (South Italy), *Tectonophysics, 98,* 29-41, 1983.

Van der Voo, R., *Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans,* 405 pp., Cambridge Univ. Press, New York, 1993.

C. Kissel and F. Speranza, Centre des Faibles Radioactivités, Laboratoire mixte CNRS/CEA, Avenue de la Terrasse, 91198 Gif-sur-Yvette Cédex, France. e-mail: kissel@cfr.cnrs-gif.fr

V. Milicevic, Geomagnetic Institute, 11306 Grocka, Serbia.

(Received December 12, 1994; revised March 31, 1995;
accepted April 10, 1995)

21562202b, 1995, B8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/95JB01243 by Curtin University Library, Wiley Online Library on [28/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License